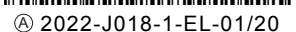


270 marks



Instructions

There are 13 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

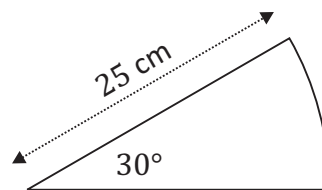
Write the make and model of your calculator(s) here:

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Question 1 (Suggested maximum time: 5 minutes)

Question 1 (Suggested maximum time: 5 minutes)

- (a) A circular pizza of radius 25 cm is cut into equal-sized slices. Each slice is in the shape of a sector of angle 30° , as shown in the diagram.



- (i) Write down the diameter of the pizza.

[illegible]

- (ii)** How many slices was the pizza cut into?

[illegible]

- (b)** A local pizzeria has a mid-week meal deal. The deal offers any pizza with a choice of base, one side order and a drink, as shown in the table below. For example, a customer can pick a classic base Margherita pizza with garlic bread and lemonade.

Pizza	Base	Side Order	Drink
Margherita (v)	Classic	Caesar Salad	Sparkling Water
Americana	Wholemeal	Garlic Bread	Still Water
Tropicana	Italian	Potato Wedges	Lemonade
Meat Feast	Thin & Crispy	Chicken Strips	Cola
Vegetable Supreme (v)		BBQ Wings	Orange
Vegan Vegetable Supreme (v)			

Note: (v) indicates a vegetarian option.

- (i) How many different meal deal choices are possible from the pizzeria?

[illegible]

- (ii) What is the probability that a person orders a pizza suitable for a vegetarian with garlic bread as part of their meal deal?

[illegible]

Question 2 (Suggested maximum time: 5 minutes)

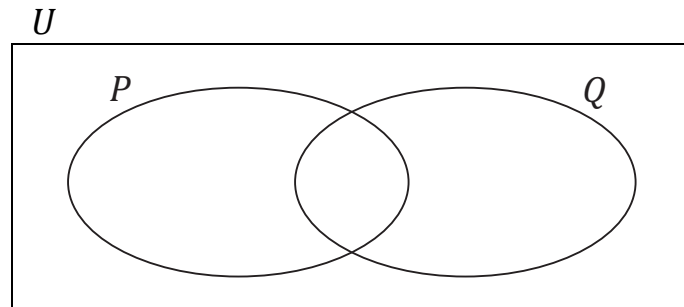
Question 2 (Suggested maximum time: 5 minutes)

- (a)** The universal set U contains set P and set Q .
The union of sets P and Q , written using set notation, is $P \cup Q$.

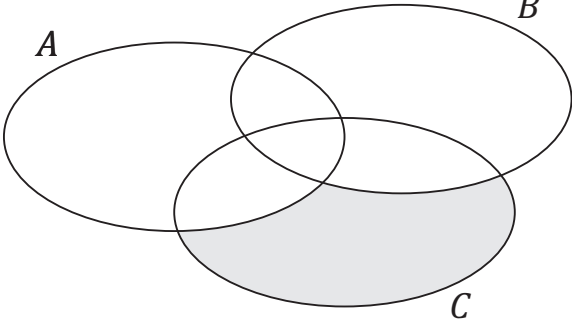
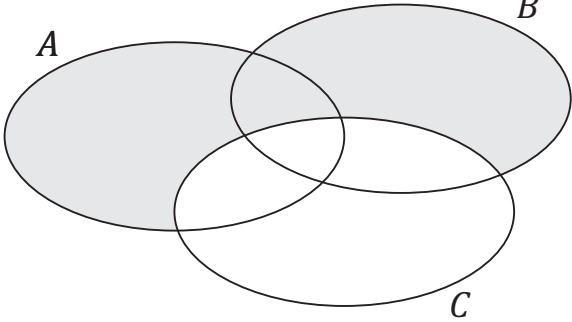
- (i)** Describe what is meant by $P \cup Q$ in your own words.

[illegible]

- (ii) **Shade** in $P \cup Q$ in the Venn diagram.



- (b)** Describe, using **set notation**, the shaded regions in each of the following Venn diagrams.

	Venn Diagram	Set Notation
(i)		
(ii)		

Question 3 (Suggested maximum time: 15 minutes)

Question 3 (Suggested maximum time: 15 minutes)

Income tax is deducted at the rate of 20% on the first €35 300 and 40% on the balance.

- (a) Joan has a gross annual income of €65 000. She has annual tax credits of €3545. Work out Joan's **net income tax** payable for the year.

[illegible]

- (b)** Joan also pays a total of €5750 in PRSI, USC and pension contributions in the ratio 10:8:5. Work out the amount of PRSI, USC and pension contributions that she pays.

PRSI = € USC = € Pension = €

[illegible]

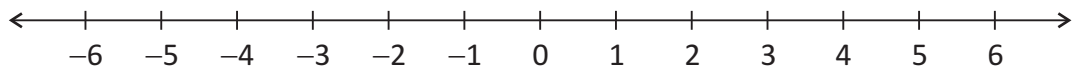
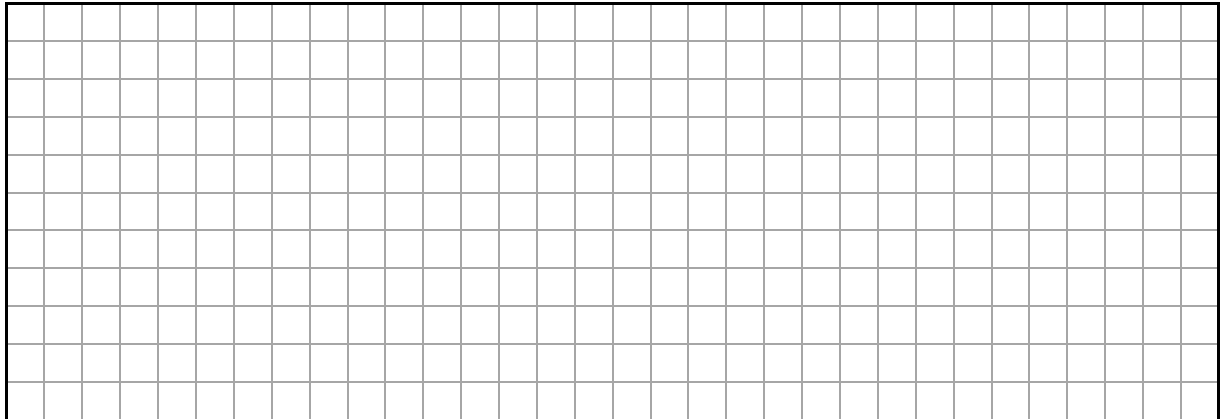
- (c) Work out Joan's **net pay** for the year.

[illegible]

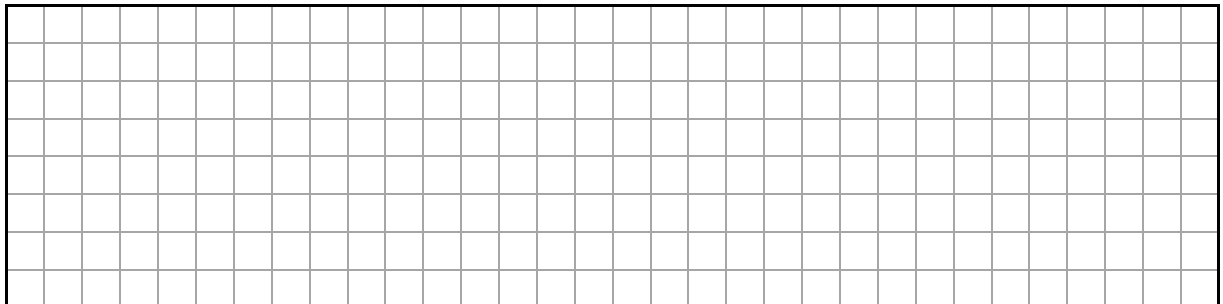
Question 4**(Suggested maximum time: 5 minutes)**

- (a) Solve the following inequality and show the solution on the number line.

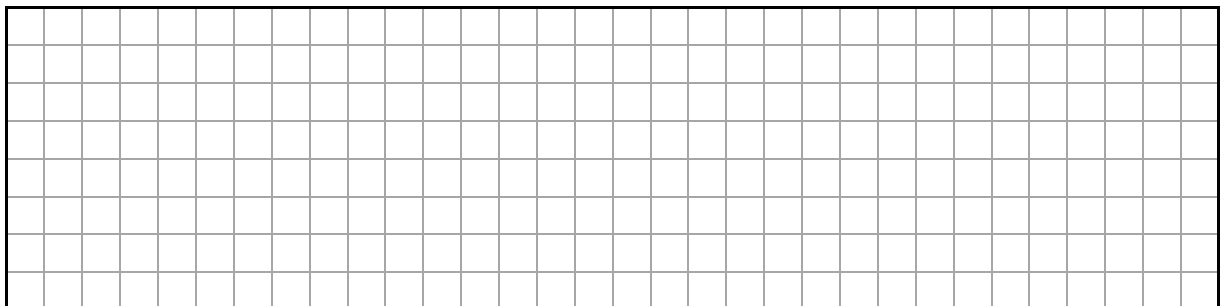
$$5x - 7 \leq 18, \quad x \in \mathbb{R}$$



- (b) Factorise fully $2rs - 4rt - 6t + 3s$.

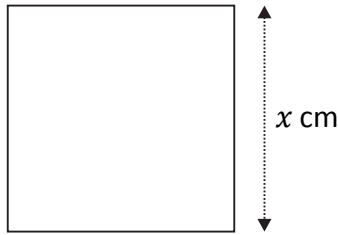


- (c) Factorise fully $5a^2 - 125b^2$.

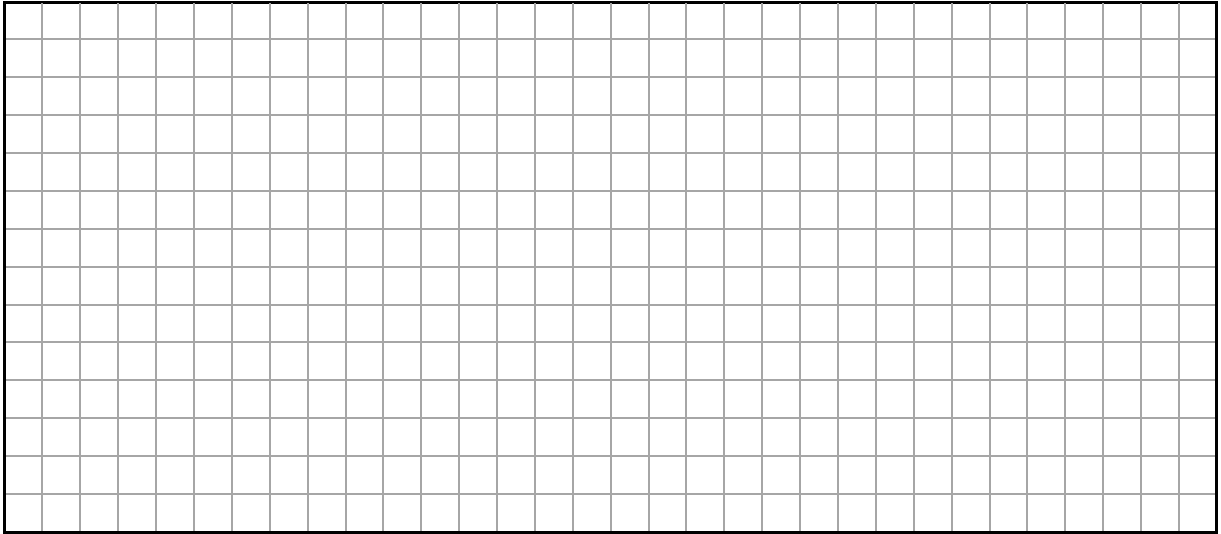


Question 5**(Suggested maximum time: 5 minutes)**

In the diagram shown, the length of a side of the square is x cm, where $x \in \mathbb{R}$.



Find the length of the diagonal of the square in terms of x .
Give your answer in surd form.



Question 6

(Suggested maximum time: 5 minutes)

Emma makes bracelets and necklaces using multi-coloured beads of different sizes.

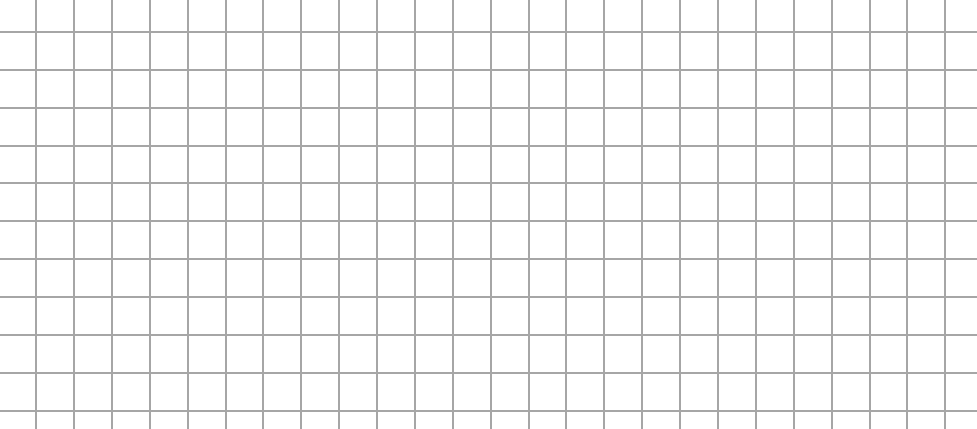
A bracelet consists of a total of n beads.

A necklace has 56 more beads than a bracelet.



- (a) Write an expression for the total number of beads used to make a bracelet and a necklace.

- (b)** Emma used 100 beads to make a bracelet and a necklace. Find the number of beads used to make the bracelet.

A large grid of graph paper, consisting of 20 columns and 20 rows of squares, intended for drawing a picture.

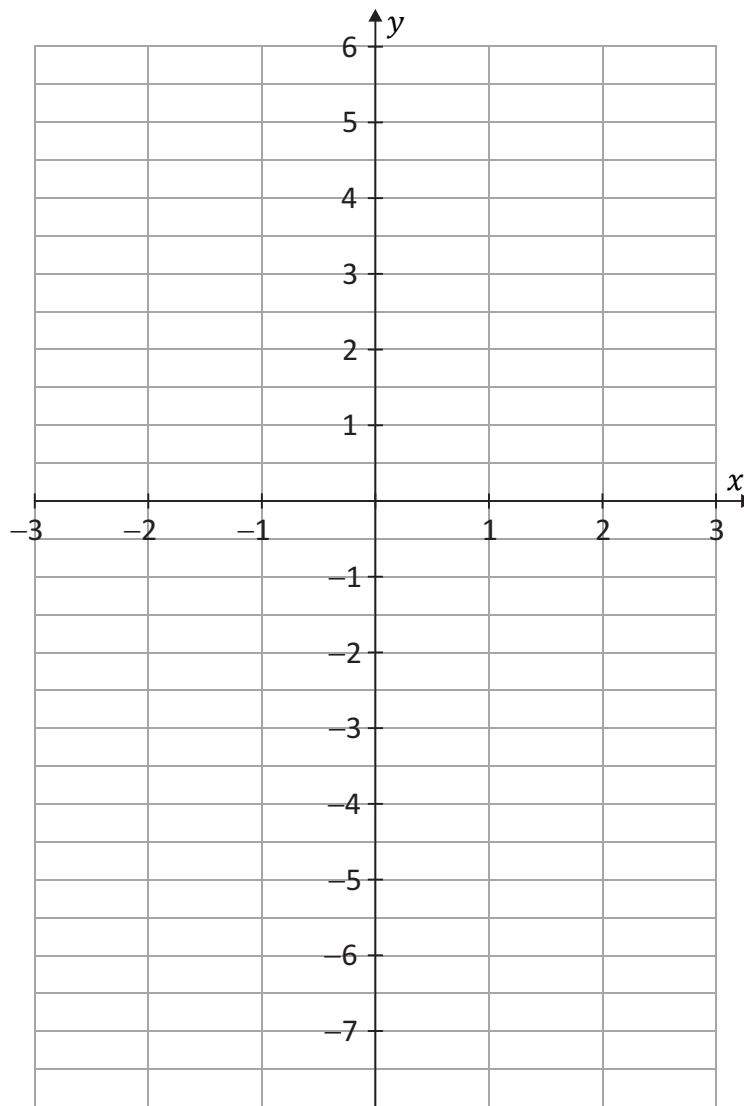
Question 7**(Suggested maximum time: 15 minutes)**

A function is defined as $f(x) = 2x^2 - x - 6$, where $x \in \mathbb{R}$.

- (a) Write down the two factors of $f(x)$.

- (b) Hence, or otherwise, solve for $f(x) = 0$.

- (c) **Draw** the graph of the function $f(x) = 2x^2 - x - 6$ in the domain $-2 \leq x \leq 2$, where $x \in \mathbb{R}$, on the co-ordinate grid opposite. Show your working out.



- (d)** Use your graph to show that your answers from part **(b)** are correct.

[illegible]

- (e) On your graph, show the **axis of symmetry** of the function $f(x) = 2x^2 - x - 6$. Write down the equation of this axis of symmetry below.

Answer =

--

[illegible]

Question 8**(Suggested maximum time: 10 minutes)**

Rugby goalposts are fitted with foam protectors, as shown, to protect players from injury. Each protector is in the shape of a cuboid that has a cylindrical section removed from its centre where the goalpost fits.

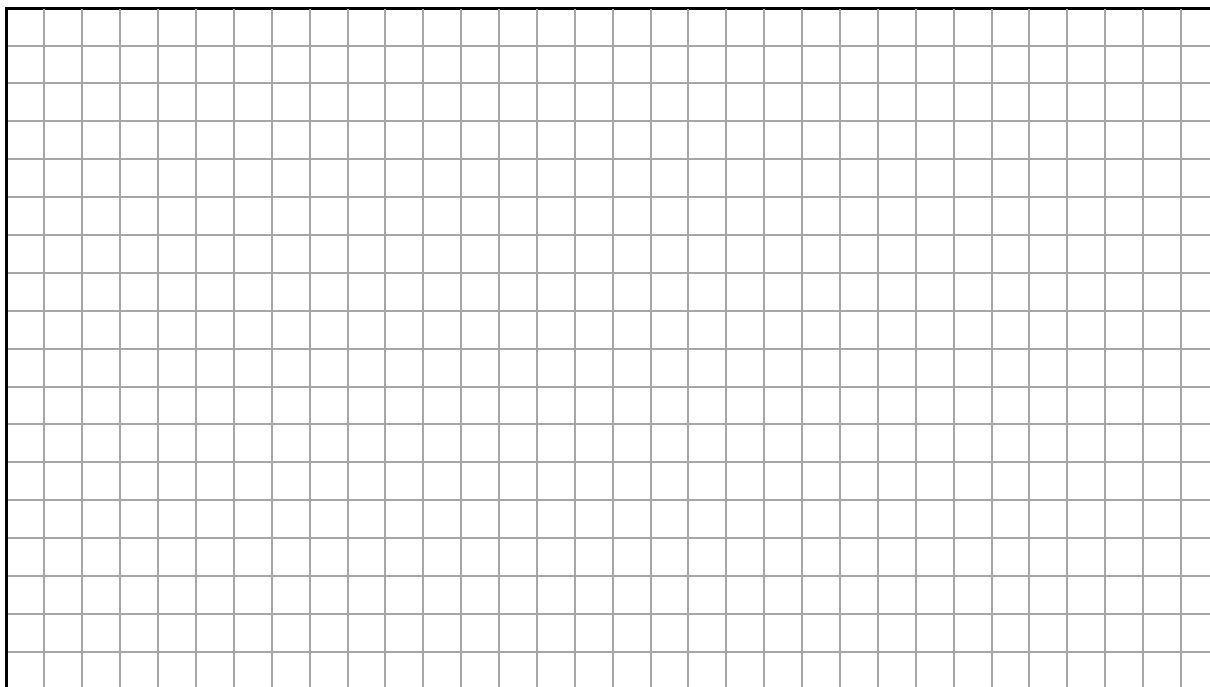
The cuboid has a square face that measures 300 mm by 300 mm and it is 1.8 m in length. The cylindrical section has a diameter of 150 mm and is also 1.8 m in length.



- (a) Find the volume of foam in one of these protectors.
Give your answer in m^3 , correct to two decimal places.



- (b) A foam protector is covered with PVC material to make it waterproof.
Find the **total** surface area of a protector which is covered by PVC material.
Give your answer in m^2 , correct to two decimal places.



Question 9

(Suggested maximum time: 5 minutes)

(a) A six-shape repeating pattern, which continues indefinitely, is shown below.



Pattern 1



Pattern 2



Pattern 3



Pattern 4

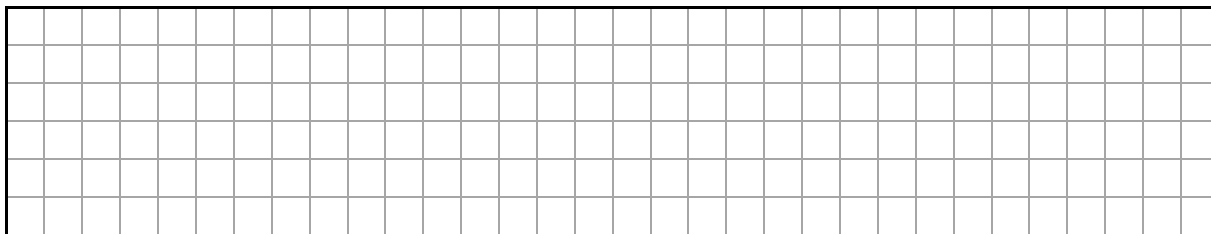


Pattern 5

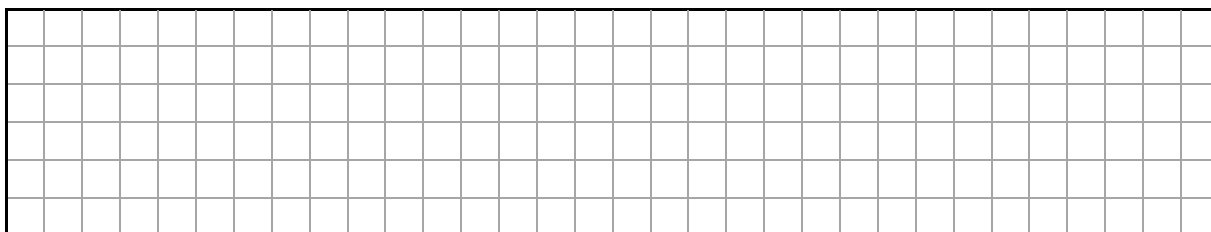


Pattern 6

(i) Work out the shape of Pattern 20.



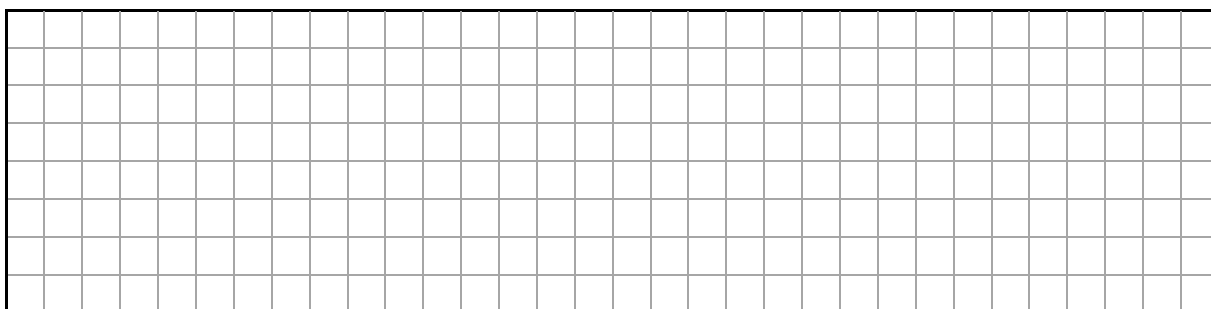
(ii) Work out the shape of Pattern 2021.



(b) In a particular **exponential** sequence, the first term is 3 and the fourth term is 81.

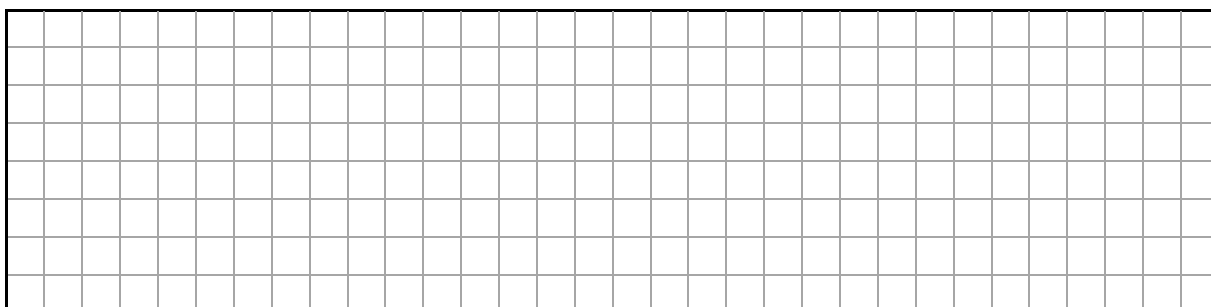
(i) Fill in the boxes below to show the first five terms of the sequence.

3, , , 81,



(ii) Show that 6561 is a term in this sequence if the pattern continues.

Hint: $\sqrt{6561} = 81$.



Question 10

(Suggested maximum time: 5 minutes)

- (a)** Brian makes the following statement:

"If a 2D shape is a square, then it must have four sides of equal length."

- (i) Complete the **converse** of Brian's statement:

If a 2D shape has four sides of equal length, then

--

- (ii) Is the **converse** of Brian's statement true or false?

Put a tick (✓) in the correct box to show your answer.

Tick **one** box only. Justify your answer.

True

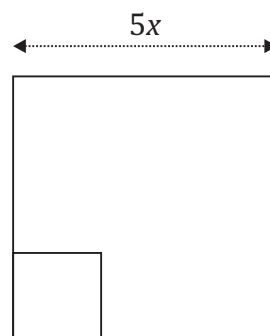
7

False

5

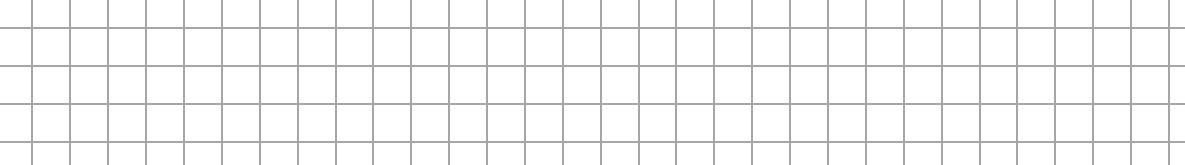
Justification:

- (b)** Two squares are shown in the diagram.
The larger square has sides of length $5x$ units.
The smaller square has an area equal to $\frac{1}{9}$
of the area of the larger square.



Find the length of the sides of the smaller square.

Give your answer in the form $\frac{a}{b}x$, where $a, b \in \mathbb{N}$.



Question 11 (Suggested maximum time: 10 minutes)

Question 11 (Suggested maximum time: 10 minutes)

Two perpendicular lines, m and n , pass through the point $(2, 3)$.

The equation of m is $y = 3x + 4$.

- (a)** Write down the **y-intercept** and the **slope** of line m .

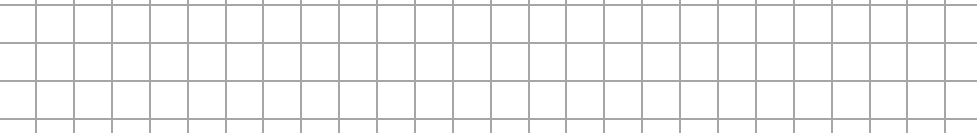
y -intercept =

slope =

[illegible]

- (b) Find the **equation** of line n .

Give your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.



- (c)** The point $(-1, 4)$ is on one of the lines, m or n .

Which line is it on? Justify your answer.

Answer =

--

- Justification:

Justification:

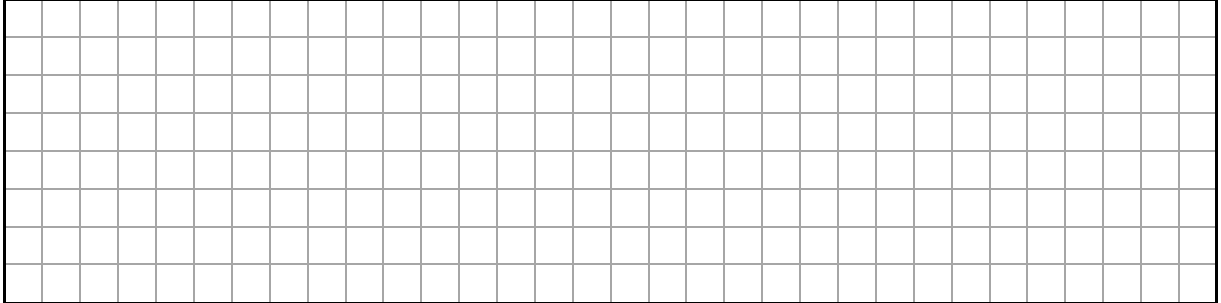
Question 12**(Suggested maximum time: 10 minutes)**

A triangle ABC has a right angle $\angle ABC$.

$|BC| = 12$ cm.

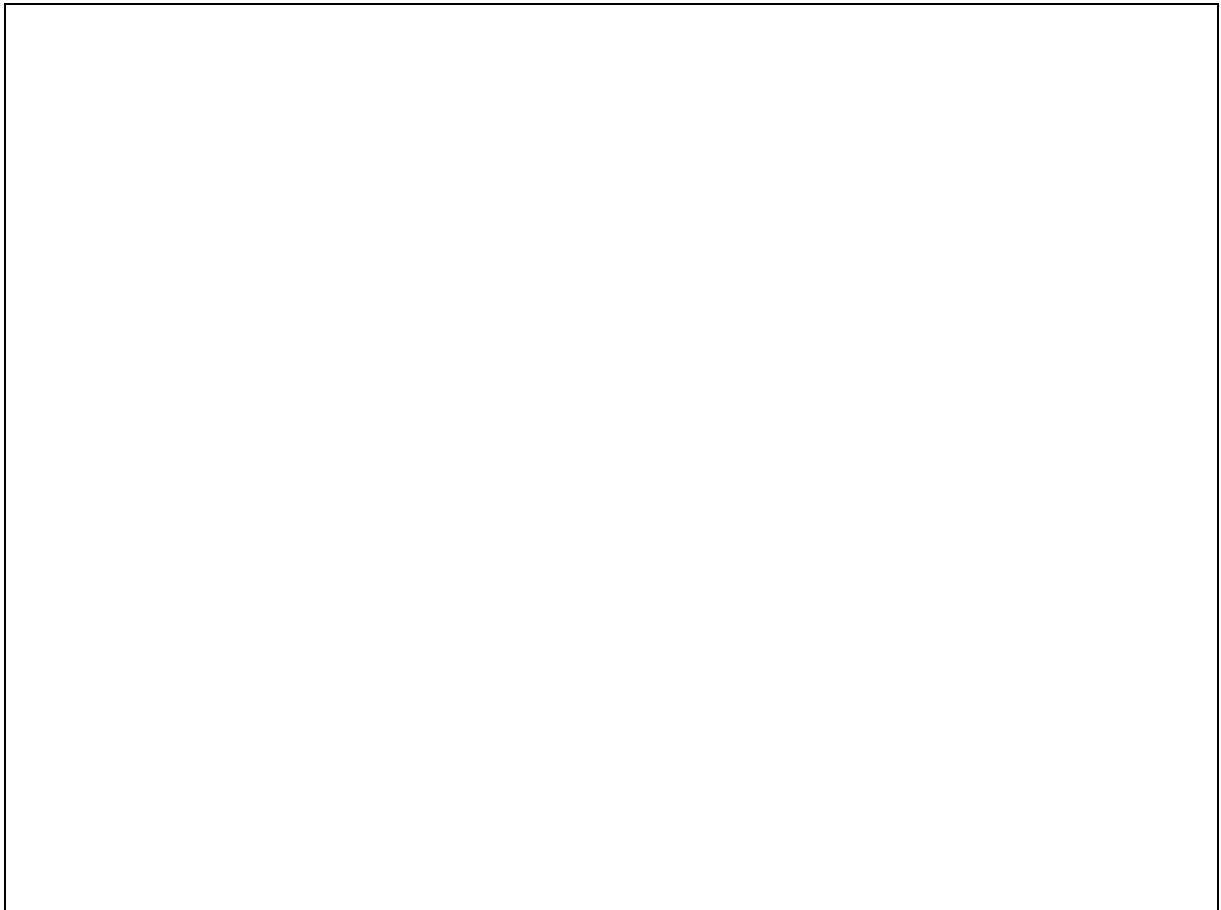
The area of the triangle is 48 cm^2 .

- (a)** Work out $|AB|$, the perpendicular height of the triangle.



- (b)** **Construct** the triangle ABC in the space below.

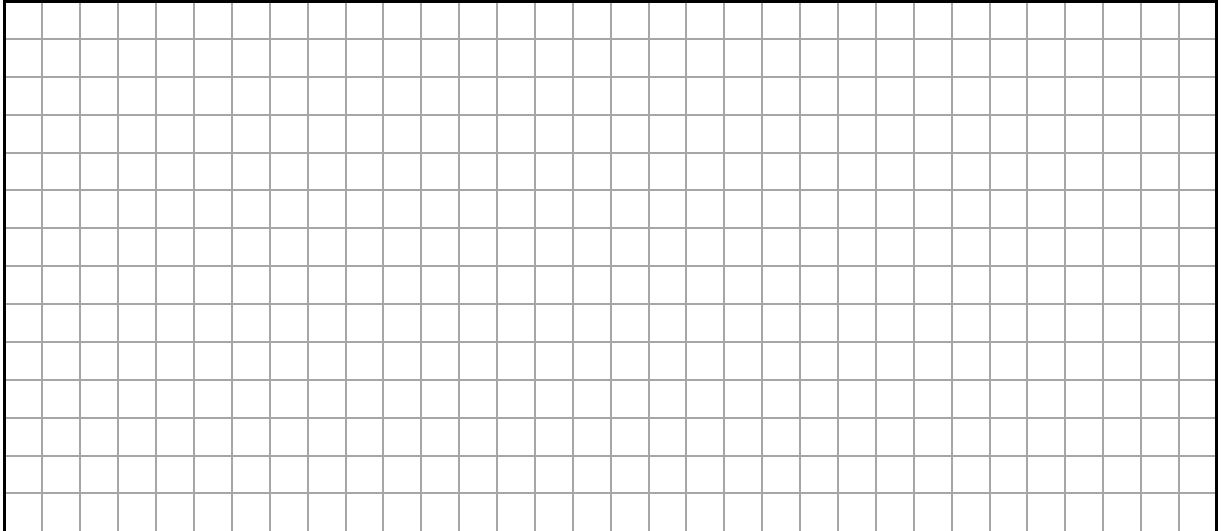
Show all of your construction lines clearly.



- (c) On your diagram, **measure** the angle $\angle BCA$.
Give your answer correct to the nearest degree.

$$|\angle BCA| =$$

- (d) Use **trigonometric ratios** to verify your answer from part (c).



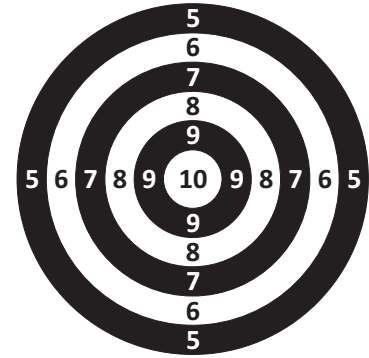
Question 13

(Suggested maximum time: 15 minutes)

In an archery competition, Liam shoots 72 arrows hitting the target shown each time.

Liam's scores are displayed in the frequency table below.

Liam's score card						
Target score	5	6	7	8	9	10
Frequency	6	11	9	14	19	x



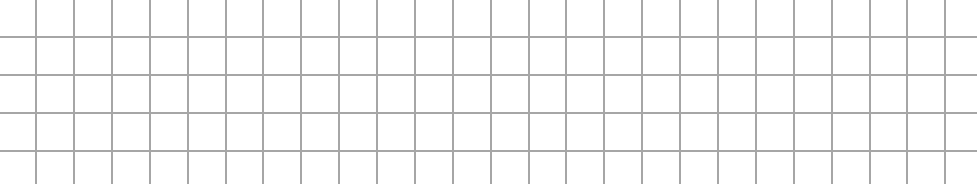
- (a)** Find x , the number of times Liam hit the bullseye (10-point zone) of the target.

A large rectangular grid consisting of 20 columns and 10 rows of small squares, intended for drawing.

- (b)** Write down the **modal** target score for Liam (the mode).

[illegible]

- (c) Find the estimated **mean** score of a single arrow, giving your answer to nearest whole number.



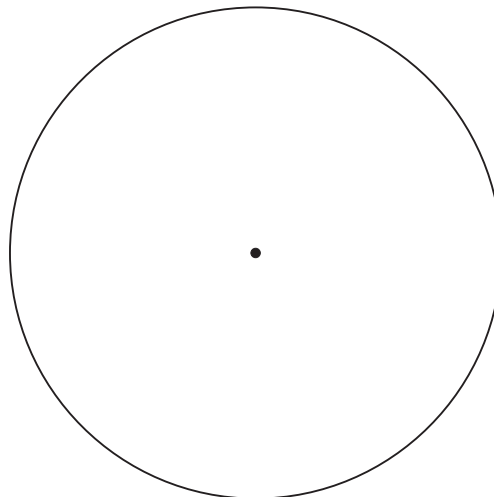
- (d)** Use the values in the table to find the **median** score. Justify your answer.

[illegible]

- (e)** In your opinion, which one of the measures of central tendency from parts **(b)**, **(c)** and **(d)** is most representative of Liam's scores? Give a reason for your answer.

Answer:	
Reason:	

- (f) Complete the pie chart below to show Liam's scores. Label each sector **and** the size of each angle clearly.

A blank sheet of graph paper featuring a uniform grid of small squares. The grid consists of 20 columns and 15 rows, providing a structured space for drawing or writing.

Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

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Pre-Junior Cycle Final Examination, 2022

Mathematics – Higher Level

Time: 2 hours

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① 2022-J018-1-EL-20/20